Monitoring Ethyl Cyanoacrylate Polymerization Kinetics by Spectroscopy

Learning Objective

In this experiment, students will investigate the polymerization kinetics of ethyl cyanoacrylate (ECA) using fluorescence and UV-Vis spectroscopy techniques. Students will apply Beer's Law and construct a pseudo-first-order kinetics plot, extracting the apparent rate constant K_{app}. Through this, students will connect the rate of anionic addition polymerization to the resulting polymer structure and polydispersity. This experiment also utilizes Nile Red as a fluorescent molecular probe, allowing students to observe changes in viscosity and the microenvironment inside the cuvette as polymerization occurs. Students will also observe why and understand how the kinetic model breaks down at high monomer conversion, and how that affects the resulting polymer structure.

Introduction

Available in any hardware store, somewhere on the shelf, you will almost certainly find superglue. It's extremely cheap, versatile, and has a variety of uses ranging from closing wounds in medical applications to quick repairs around the house, and even forensic fingerprint development. This lab aims to explain what actually happens when superglue leaves the tube, how fast it happens, and what we can learn by watching it happen.

The active ingredient is ethyl cyanoacrylate. This small, reactive molecule waits patiently until it comes into contact with water, specifically hydroxide ions. The moment it finds even a trace, a chain reaction kicks off. Molecules begin linking end to end, forming long polymer chains that lock onto whatever surface they are touching. In a few seconds, a liquid becomes a solid and two surfaces become permanently bound together.

In 1942, chemist Harry Coover was attempting to develop clear plastic gun sights for the Allied forces during World War II. While working with cyanoacrylate compounds, he noticed they were sticking to absolutely everything. When trying to manufacture precision optical equipment, this is a bit of a problem. Cooper shelved the discovery and moved on. Nearly a decade later, he finally recognized what he actually had, and by 1958, the first iteration of superglue became commercially available.

Since then, cyanoacrylate chemistry has found its way into places far removed from the hardware store. Surgeons use it to close wounds without sutures, forensic investigators use its vapor to lift fingerprints from crime scenes, and researchers are exploring it as a material for targeted drug delivery. In each case, it's the same lightning-fast polymerization reaction doing the work.

How does it actually work, and can we slow it down enough to watch it? That is what this experiment aims to answer. Using UV-Vis absorption and fluorescence spectroscopy, we will track the polymerization of ethyl cyanoacrylate in real time, measuring how fast the monomer disappears and watching the liquid transform into a solid right inside the cuvette.

Theory

Ethyl cyanoacrylate (ECA) is a small organic molecule in the acrylate family. What differentiates ECA from other acrylates is the presence of two strong electron-withdrawing groups flanking the vinyl group: an ethyl ester group ($COOC_2H_5$) and a nitrile group (CN). Together, these groups pull electron density away from the carbon-carbon double bond, leaving the β-carbon highly susceptible to attack by nucleophiles. This is why ECA is so reactive. It does not need UV light, elevated temperature, or a radical initiator to polymerize. A nucleophile will work, even a weak one.

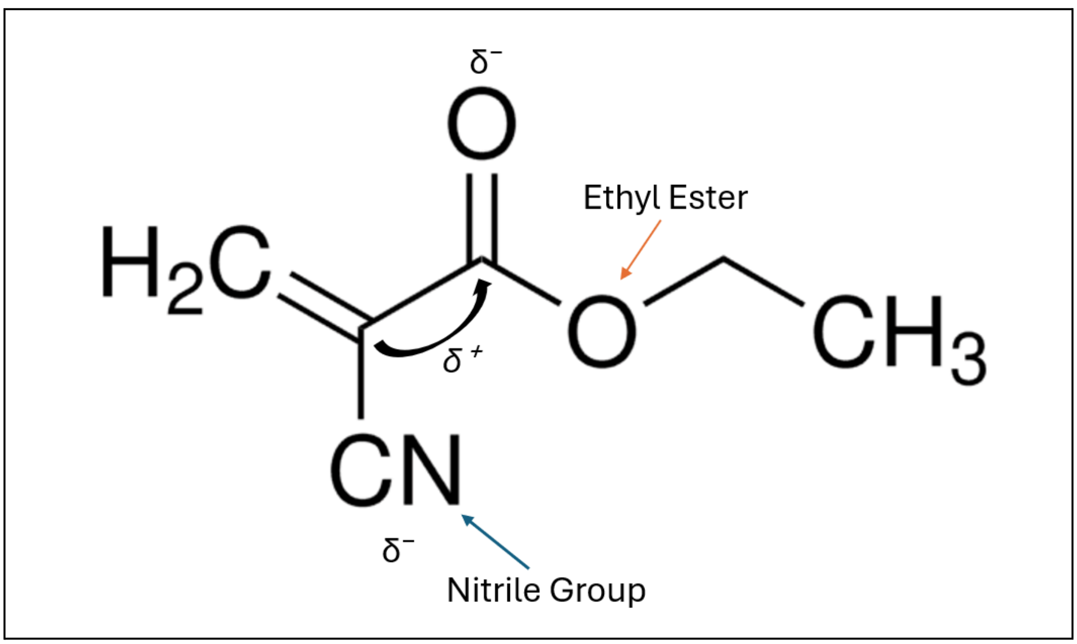

Figure 1: Structure of an ECA monomer with the nitrile and ethyl ester group labeled, electron-withdrawing nature indicated by arrows.

When a nucleophile, most commonly a hydroxide ion, encounters ECA, it attacks the electron-deficient β-carbon on the vinyl group, initiating polymerization. This first step generates a carbanion at the α-carbon. Rather than terminating, the carbanion is stabilized through resonance delocalization across the nitrile and ester groups flanking it. This stability is what allows the chain to keep growing, as the carbanion lives long enough to encounter another monomer molecule and continue the reaction.

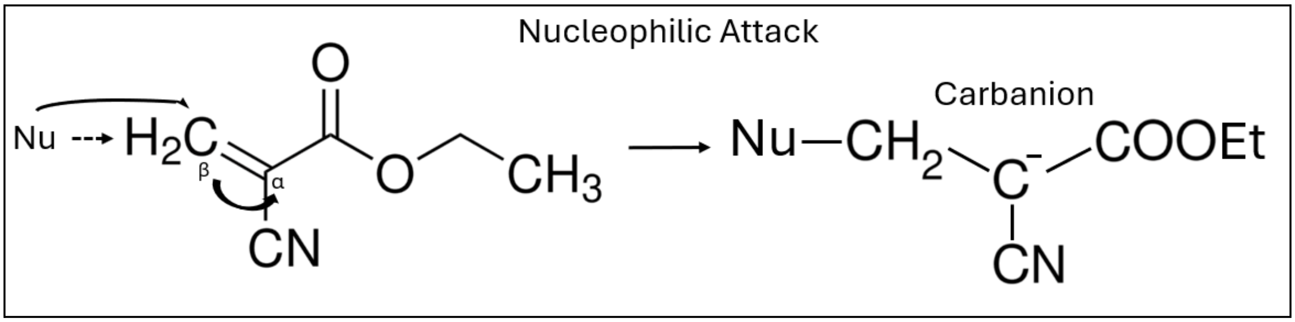

Figure 2: Initiation step showing nucleophilic attack on the β-carbon and the resulting carbanion. (**Note:** There are three resonance structures in total of the resulting carbanion, only one of which is shown.)

After initiation, each active chain end rapidly adds monomers in a process called propagation. Each addition regenerates a stabilized carbanion at the chain terminus, which immediately attacks the nearest available monomer. This process repeats until the chain encounters something that quenches the carbanion, most commonly a proton source or cationic impurity, at which point termination occurs, and chain growth stops.

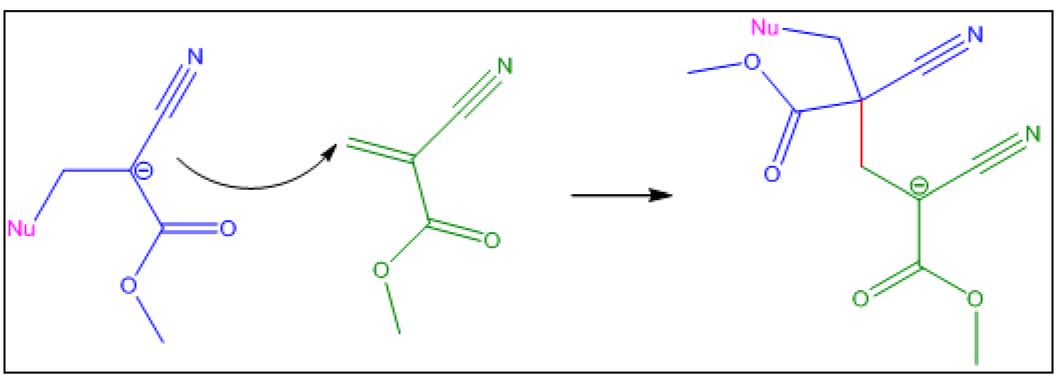

Figure 3: Propagation of ECA showing the carbanion acting as the new nucleophile attacking a second monomer; this continues until termination.

Now that we have established the mechanism, we can describe the rate at which polymerization occurs. Since initiation by hydroxide ions is fast relative to propagation, it can be assumed that all initiator molecules generate active chain ends almost immediately upon contact. This means concentration of active chain ends remains constant throughout propagation, and is equal to the initial initiator concentration $[I]_0$. Under these conditions, the rate of monomer consumption simplifies to:

$$\frac{-d[M]}{dt} = k_p[I]_0[M]$$

Since $[I]$ is constant throughout the reaction, it can be combined into the single apparent rate constant, k_{app}:

$$k_{app} = k_p[I]_0$$

This gives us a pseudo-first-order expression:

$$\frac{-d[M]}{dt} = k_{app}[M]$$

Integrating this expression with respect to time yields a linear relationship:

$$ln(\frac{[M]_0}{[M]_t}) = k_{app}t$$

This is the key expression for this experiment. If polymerization follows ideal pseudo-first-order kinetics, a plot of $ln(\frac{[M]_0}{[M]_t})$ versus time will produce a straight line, with the slope equal to k_{app}.

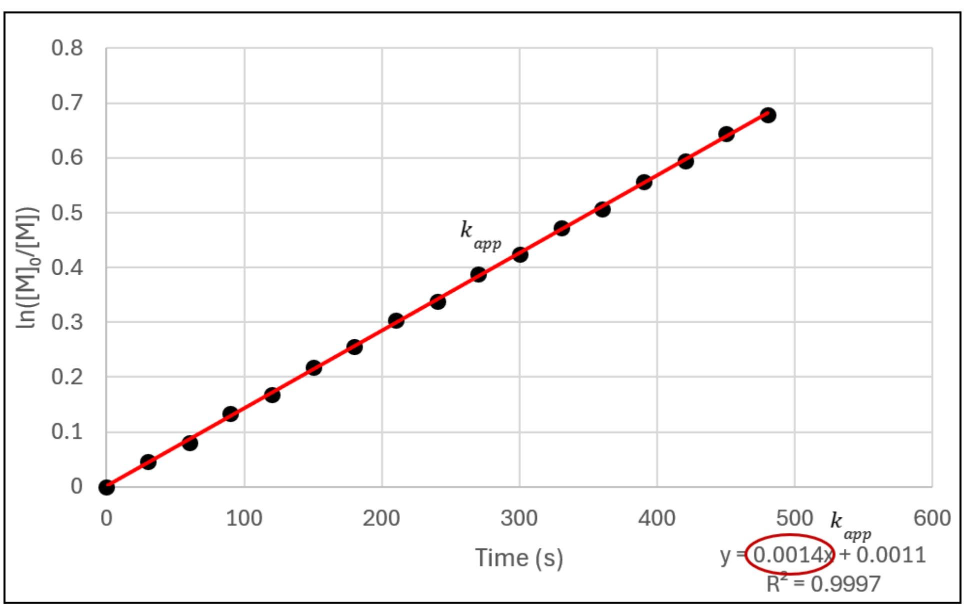

Figure 4: Example of a pseudo-first-order kinetic plot with slope labeled as k_{app}.

While ECA polymerization follows this pseudo-first-order kinetics model at the start of the reaction, it does not remain ideal. As monomer is consumed, polymer chains accumulate, and the solution becomes increasingly viscous. Eventually, the viscosity becomes high enough that diffusion of monomer to active chain ends slows significantly. The rate of termination, which depends on two chain ends finding each other, falls more quickly than the rate of propagation. This is because propagation only requires a chain end to find a monomer, which is plentiful, while termination requires two chain ends to find each other in a crowded environment. The result is that the reaction counterintuitively speeds up before eventually slowing as the monomer is depleted. The rate increases due to reduced termination, not because propagation is intrinsically faster. This phenomenon is known as the Trommsdorff-Norrish effect, also called the gel effect, and it can lead to dangerous consequences when working on a large scale, as polymerization is commonly an exothermic process.

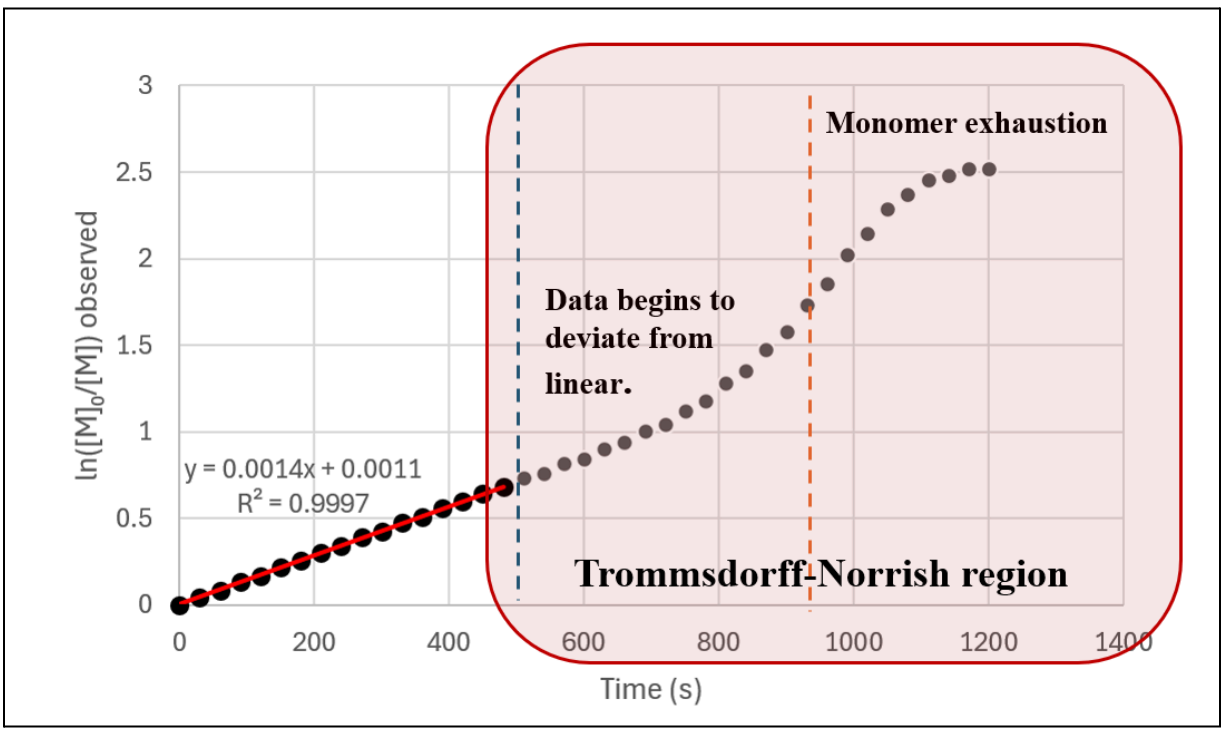

Figure 5: Kinetic plot showing deviation from linearity at high conversion with the Trommsdorff-Norrish region labeled.

The Trommsdorff-Norrish effect has direct consequences for the ECA polymerization reaction. In reality, initiation will start unevenly across the sample as the nucleophile cannot be introduced in a perfectly uniform manner. This means chains are started at slightly different times, resulting in a broad distribution of chain lengths in the final polymer. This is quantified by the polydispersity index (PDI) and can be defined as:

$$PDI = \frac{M_w}{M_n}$$

Where M_w is the weight average molecular weight, a measure of average molar mass that favors longer chains, which can be calculated as:

$$M_w = \frac{\Sigma N_i M_i^2}{\Sigma N_i M_i}$$

And M_n is the number average molecular weight, or the total mass of the polymer sample divided by the total number of chains.

$$M_n = \frac{\Sigma N_i M_i}{\Sigma N_i}$$

Radical polymerization yields a PDI value ranging from 1.5 to 2, due to initiation, propagation, and termination all occurring in unison. On the other hand, ideal polymerization approaches PDI = 1 since all chains start nearly simultaneously and termination is suppressed. In a perfectly uniform polymer where all chains are identical, PDI equals 1. The broader the distribution of chain lengths, the higher the PDI. For an ideal anionic polymerization with fast initiation and no termination, PDI approaches 1. ECA deviates from this ideal in two key ways. First, initiation is not perfectly instantaneous, meaning chains begin growing at slightly different times. Second, the Trommsdorff-Norrish effect broadens the distribution further at high conversion. The result is a PDI that is above 1, despite the anionic mechanism being more controlled than radical polymerization.

We can now discuss the analytical tools used to monitor the polymerization in real time. The primary tool in Part A of this experiment is UV-Vis spectroscopy. Beer's Law describes the relationship between absorbance and concentration:

$$A = \varepsilon l c$$

In which A is the absorbance, ε is the molar absorptivity, l is the path length of the cuvette, and c is the concentration of the analyte. ECA absorbs UV light at 270 nm due to a $\pi \rightarrow \pi^*$ transition, where an electron is being promoted from the bonding π orbital to an antibonding π^* orbital. This transition is made possible by the conjugation located between the vinyl group and the two electron-withdrawing groups that flank it. As the polymerization reaction advances, the vinyl group is consumed, and as a result, the absorbance at 270nm decreases proportionally. By applying Beer's Law, a monomer concentration can be calculated at each point in time, allowing for the construction of a pseudo-first-order kinetic plot.

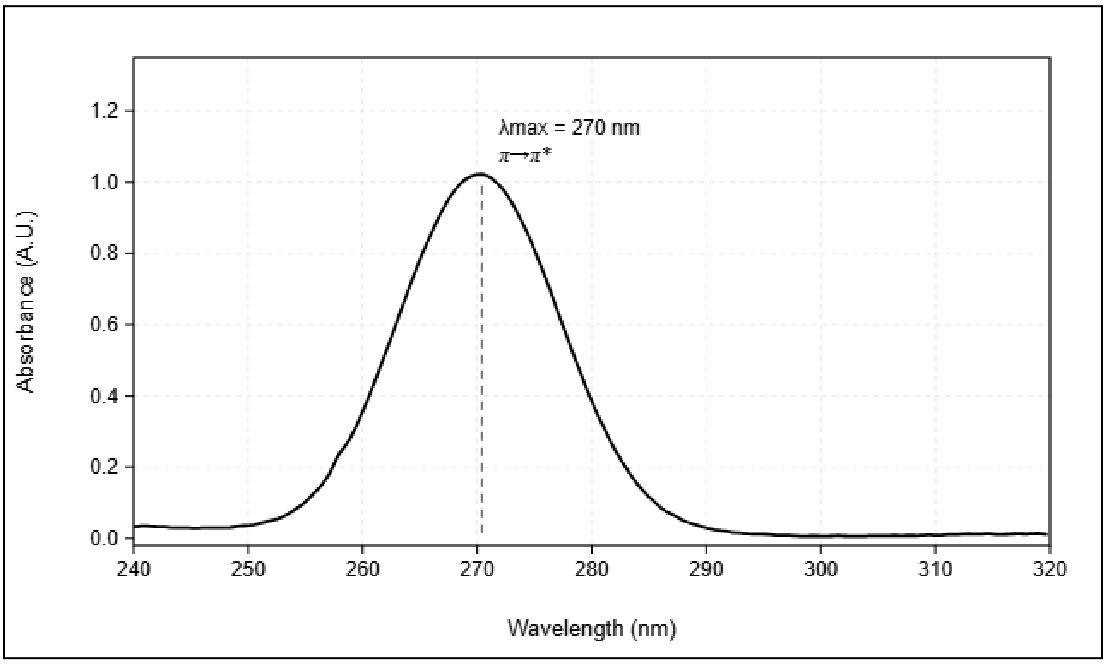

Figure 6: Theoretical UV-Vis absorption spectrum of ECA monomer showing a peak at 270nm

The second analytical tool employed in this lab is fluorescence spectroscopy. Nile Red is a hydrophobic fluorescent dye that is highly sensitive to the polarity of its surrounding environment. This property is known as solvatochromism. In a polar environment, Nile Red emits at a longer wavelength, while in a nonpolar or a viscous environment, as seen at the end of the polymerization reaction, its emission blue shifts to a shorter wavelength. Before polymerization, Nile Red is dissolved in the ECA monomer solution and emits at a wavelength that reflects the polar environment. As polymerization is initiated, the environment becomes increasingly viscous and non-polar, as polymer chains surround the dye molecules. The change in the environment causes a shift in the Nile Red emission spectrum, providing real-time polymerization progress to complement the UV-Vis data.

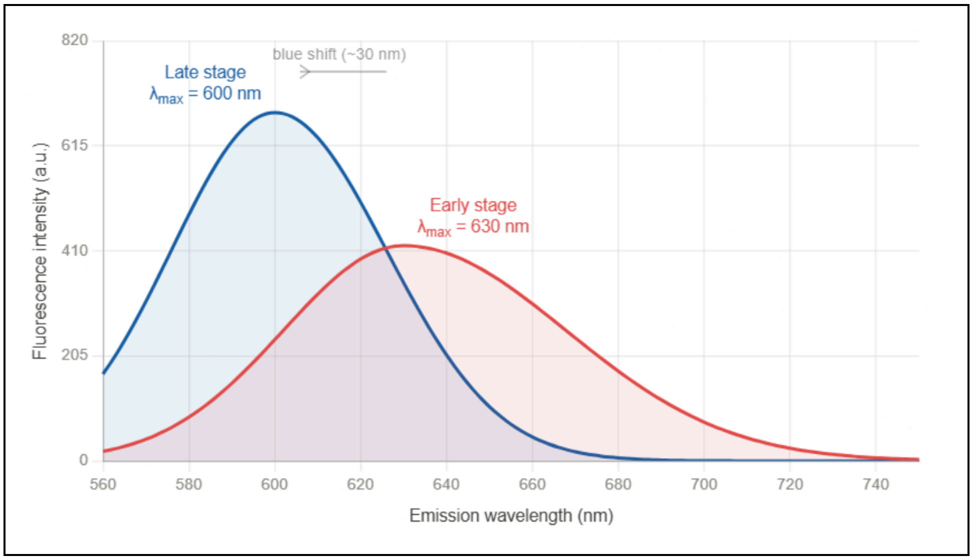

Figure 7: Theoretical Nile Red emission spectrum showing blue shift from early to late stage polymerization

While UV-Vis and fluorescence spectroscopy are the primary analytical techniques used in this experiment, proton NMR spectroscopy provides a different view of ECA polymerization. In the ECA monomer, the two vinyl protons show distinct peaks in the ^{1}H NMR spectrum. As polymerization proceeds, the signals disappear and are replaced by signals of the polymer backbone. Through tracking the relative intensity of the vinyl protons over time against a standard, the monomer conversion can be quantified. Although we are not running an NMR in this experiment, sample NMR data is provided to illustrate how monomer consumption can be tracked and complement the kinetic data from the UV-Vis spectroscopy.

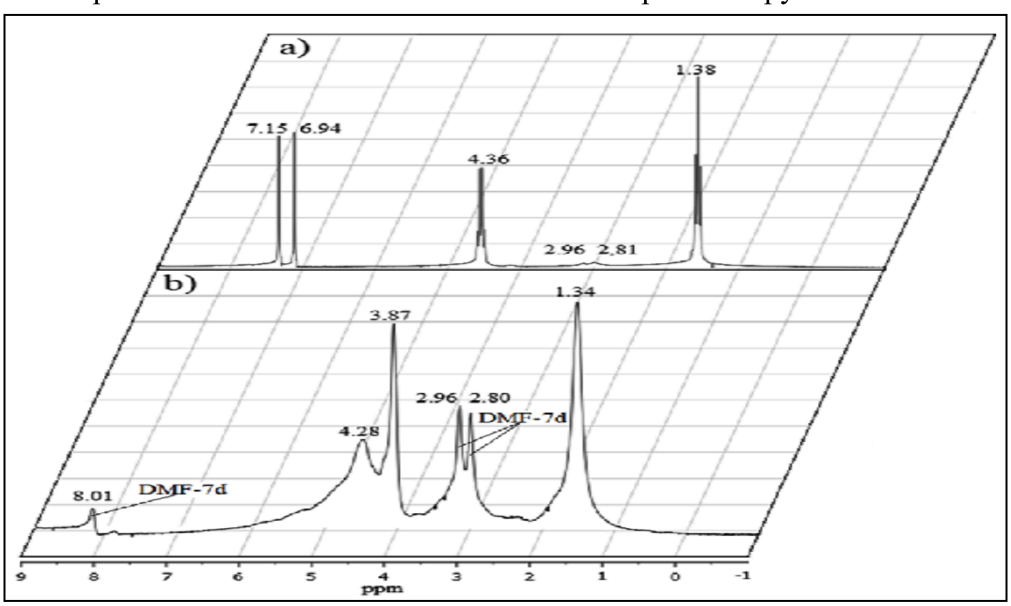

Figure 8: Sample ^{1}H NMR spectra of ECA monomer and polymer showing vinyl proton before and after polymerization.

SAFETY INFORMATION

All students must wear safety goggles, a laboratory coat, and nitrile gloves throughout the entirety of this experiment.
Ethyl cyanoacrylate is the most significant hazard in this experiment. It can cause skin irritation, eye irritation, or respiratory irritation. It will polymerize on contact with any moisture, including skin. In the event of skin bonding, soak the area in warm soapy water and gently peel apart with a blunt instrument.
Acetone is a flammable solvent and must be kept away from flames and ignition sources at all times. Vapors can irritate the eyes and respiratory tract.
Nile Red is a staining dye and will permanently stain skin, clothing, and bench surfaces.
Sodium bicarbonate does not present any significant hazards in the experiment.

Waste Disposal

Polymerized ECA waste, including cuvettes, should be disposed of in an appropriate chemical waste container. Do not put ECA or ECA-containing solutions into drainage systems. Acetone waste should be collected in a designated waste container.

Chemicals and Reagents

Chemical	Formula	M/W (g/mol)	Amount/ Concentration	Role
Ethyl Cyanoacrylate (Loctite Super Glue or >95% ECA equivalent)	$C_6H_7NO_2$	125.13	1 drop (~50 µL) per cuvette	Monomer
Sodium Bicarbonate	$NaHCO_3$	84.01	1.5 mM aqueous stock, 10 µL added per cuvette (5 µM final concentration)	Initiator
Nile Red	$C_{20}H_{18}N_2O_2$	318.37	1 mM stock in dry acetone (dissolve 0.318 mg of Nile Red per mL of dry acetone) 2 µL per cuvette	Fluorescent Probe
Acetone (Dry)	C_3H_6O	58.08	3 mL per cuvette (Part A), 2.5 mL per cuvette (Part B), chilled to -20°C	Solvent

Table 1: Chemical reagents, M/W (g/mol), amount/concentration for experiment and role.

Experimental Procedure

Part A - UV-Vis Spectroscopy

1. Prepare a 15 mM $NaHCO_3$ stock solution by dissolving 12.6 mg of $NaHCO_3$ in 10mL of distilled water. Using a micropipette, transfer 1 mL of this stock into a 10mL volumetric flask and dilute to the mark with distilled water to give a concentration of 1.5 mM $NaHCO_3$. Set aside.
2. Set up the UV-Vis spectrometer in the range of 240-320 nm. Allow the spectrometer to set up and warm for approximately 5 minutes.
3. Transfer 3 mL of chilled dry acetone into a fresh UV-grade cuvette. Blank the spectrometer using dry acetone. (Dry acetone is used to prevent trace water from prematurely initiating the polymerization before the addition of the initiator.)
4. Working quickly, pierce the tip of the superglue tube if not already open, and dispense one drop of ECA directly into the cuvette. Swirl once to get it to dissolve. One drop is approximately 50 µL. Calculate the approximate concentration of ECA.
5. Place the cuvette in the spectrometer and record the initial spectrum. Note the absorbance at 270 nm A_0. Set up the spectrometer to record a full spectrum every 30 seconds.
6. Add 10 µL of the 1.5 mM $NaHCO_3$ to the cuvette using a micropipette. Mix with a gentle swirl and immediately put the cuvette back into the spectrometer.
7. Begin acquisition immediately, recording a full spectrum every 30 seconds.
8. Continue recording the spectrum for 20 minutes or until gelation is observed. Note any visible changes in viscosity or cloudiness.
9. Dispose of the cuvette in the appropriate waste containers.

Part B - Fluorescence Spectroscopy

1. Transfer 2.5 mL of chilled dry acetone into a clean disposable cuvette. Working quickly, dispense one drop of ECA into the cuvette, swirl to dissolve.
2. Add 2 µL of the 1 mM Nile Red stock solution to the cuvette using a micropipette and swirl once.
3. Set up the fluorescence spectrometer with an excitation wavelength of 550nm and an emission scan range of 560-750 nm.
4. Place the cuvette in the spectrometer and record the emission spectrum for t = 0.
5. Set the fluorescence spectrometer to record a full emission spectrum every 20 seconds.
6. Begin recording the spectrum. At t = 60, remove the cuvette from the spectrometer and add 10 µL of the 1.5 mM $NaHCO_3$ solution prepared in Part A using a micropipette. Mix with a single pipette stroke and return to the spectrometer.
7. Continue acquisition of the spectrum for 30 minutes without opening the instrument. Note any visible changes in the solution at the end of the experiment.
8. Dispose of the used cuvette in the appropriate waste containers.

Part C - NMR Data Interpretation

1. Examine the two ^{1}H NMR spectra provided in **Figure 8.** Using the chemical shift reference table in the supplemental information, assign each labeled peak in both spectra to the corresponding proton environment in the ECA structure.
2. Identify which spectrum corresponds to the monomer and which corresponds to the poly(ECA). Justify your identification.
3. Locate the region between 6.5 and 7.5 ppm in both of the spectra. Comment on what the presence or absence of signals in this region indicates about the vinyl group of each sample.
4. Based on your peak assignments, identify which signals you would monitor over time to track the progress of polymerization, and explain why.

Data Analysis

From the UV-Vis spectrum collected in Part A, record the absorbance at 270 nm for each time point. Using Beer's Law and the Initial absorbance A_0, calculate the monomer concentration at each time point. Construct a plot of $ln(\frac{[M]_0}{[M]_t})$ versus time. In the linear regression, the slope of the linear region gives the apparent rate constant k_{app}. Note where the plot begins to deviate from a linear graph; this is the onset of the Trommsdorff-Norrish effect.

From the fluorescence spectrum collected in Part B, track the wavelength of maximum emission for each time point. Plot emission versus time and note the magnitude and direction of the shift over the course of the experiment.

Discussion Questions

1. From the pseudo-first-order kinetic plot you created, report your experimentally determined k_{app}. Was your plot linear throughout the entire experiment? If not, where does it deviate, and what does this indicate?
2. Based on the conditions of this experiment, would you expect the PDI of your poly(ECA) sample to be closer to 1 or significantly greater than 1? Justify your answer with factors observed in the experiment. What instrument would you need to actually measure PDI?
3. Describe the shift in the Nile Red emission wavelength observed over part B. What does the shift tell you about the microenvironment inside the cuvette and how it changed during polymerization? How does this relate to the kinetic data from Part A?
4. This experiment monitored the polymerization of ECA using three analytical techniques: UV-Vis, Fluorescence, and proton NMR spectroscopy. Each technique looks at a different aspect of the same polymerization reaction. What do each of these techniques measure? What are the advantages of fusing multiple analytical approaches to study a single chemical process?

Bibliography

1. Baig, S. Cyanoacrylate (Superglue). *Molecule of the Month*, July 2009. University of Bristol. https://www.chm.bris.ac.uk/motm/superglue/superglueh.htm (accessed 2026). DOI: 10.6084/m9.figshare.5371492.
2. Lemelson-MIT Program. Harry Coover. https://lemelson.mit.edu/resources/harry-coover (accessed 2025).
3. Paiva, D. F. F.; Mafra, M. A. T.; Dos Santos, V. A. B.; Figueroba, S. R.; Dantas, A. C. G. C.; Amorim, K. S.; Neto, F. H.; Silva, C. B. D.; Groppo, F. C. Development of a Drug Delivery System Using a Compound Based on Ethyl Cyanoacrylate and Hancornia speciosa (Gomes) in a Rat Calvaria Model. *Pharmaceuticals* **2025**, *18* (11), 1695. DOI: 10.3390/ph18111695.
4. Kinetics of Step-Growth Polymerization. In *Polymer Chemistry*; Schaller, C., Ed.; LibreTexts: Davis, CA. https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Polymer_Chemistry_(Schaller)/03%3A_Kinetics_and_Thermodynamics_of_Polymerization/3.02%3A_Kinetics_of_Step-Growth_Polymerization (accessed 2026).
5. Samuel, H. S.; Nweke-Maraizu, U.; Etim, E. E. Cyanoacrylate Chemistry and Polymerization Mechanisms. *Phys. Chem. Biochem. Res.* **2024**. DOI: 10.48309/pcbr.2024.422869.1304.
6. *Polymers* 2020, 12 (9), 2011. DOI: 10.3390/polym12092011.
7. Tulig, T. J.; Tirrell, M. Toward a Molecular Theory of the Trommsdorff Effect. *Macromolecules* **1981**, *14* (6), 1501–1511. DOI: 10.1021/ma50006a070.
8. *Coatings* 2022, 12 (4), 490. DOI: 10.3390/coatings12040490.
9. Polydispersity - an overview. ScienceDirect Topics. https://www.sciencedirect.com/topics/engineering/polydispersity (accessed 2025).
10. Wang, F. W.; Sauder, D. G. Examination of Fluorescent Molecules as in situ Probes of Polymerization Reactions. In *Polymeric Drugs & Drug Delivery Systems*; 2001.
11. *J. Chem. Educ.* https://pubs.acs.org/journal/jceda8 (accessed 2026).
12. Polymer Analysis by NMR; Sigma-Aldrich Technical Document. https://www.sigmaaldrich.com/deepweb/assets/sigmaaldrich/product/documents/101/343/polymer-analysis-by-nmr.pdf (accessed 2026).

13. The Beer-Lambert Law. In *Physical and Theoretical Chemistry Textbook Maps*; LibreTexts: Davis, CA. https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Spectroscopy/Electronic_Spectroscopy/Electronic_Spectroscopy_Basics/The_Beer-Lambert_Law (accessed 2026).
14. Comparison of ^{1}H NMR spectra of the starting ethyl 2-cyanoacrylate (a) with the obtained... ResearchGate. https://www.researchgate.net/figure/Comparison-of-1-H-NMR-spectra-of-the-starting-ethyl-2-cyanoacrylate-a-with-the-obtained_fig3_357600460 (accessed 2026).
15. Ethyl 2-cyanoacrylate MSDS. Chemical Book. https://www.chemicalbook.com/msds/ethyl-2-cyanoacrylate.pdf (accessed 2026).
16. PhotochemCAD Spectra. https://omlc.org/spectra/PhotochemCAD/html/007.html (accessed 2026).
17. Nile Red Product Information; Thermo Fisher Scientific. https://assets.thermofisher.com/DirectWebViewer/private/document.aspx?prd=ACR41571 (accessed 2026).
18. Brunning, A. Analytical Chemistry – A Guide to Proton Nuclear Magnetic Resonance (NMR). *Compound Interest*, February 24, 2015. https://www.compoundchem.com/2015/02/24/proton-nmr/ (accessed April 21, 2026).

Supplemental Information